

Hui Zhong

CONTACT INFORMATION

Department of Electrical and Computer Engineering
University of Houston
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RESEARCH INTEREST

- Security and Privacy
- Quantum Computing
- Quantum Network
- Biometric Authentication for Wearable Devices

EDUCATION

University of Houston

Sept. 2022 – June. 2026

Ph.D. in Electrical and Computer Engineering

Houston, TX, USA

- Advisor: Prof. Miao Pan
- Relevant courses: Machine Learning, Intermediate Electromagnetics, Introduction to Cybersecurity, Signal Detection, Digital Signal Processing, Advanced Telecommunications, Big Data, Advanced Linear Optimization

Dalian Maritime University

Sept. 2018 – June. 2022

Bachelor in Internet of Things

Dalian, China

- Relevant courses: Computer Control System, Principle of Digital Communication, Wireless Network Technology, Embedded System, Sensor Principle and Technology, Fundamentals of Computer Networks, High-Frequency Electronic Circuits

Loughborough University

Sept. 2021 – June. 2022

Academic Exchange Student in Computer Science

Loughborough, UK

- Relevant courses: Computer Graphics, Human-Computer Interaction, Data Mining, Machine Learning

Utah Valley University

Jan. 2019 – Feb. 2019

Visiting Student in Economics & Finance

Orem, UT, USA

- Short-term visiting program focusing on Economics and Finance topics

PUBLICATIONS

Journal

- **Hui Zhong**, Keyi Ju, Xinyue Zhang, Xiaoqi Qin, Xin Fu, and Miao Pan. “Quantum Computing Catalyzes Future Quantum Networks: Efficiency and Inherent Privacy.” *IEEE Network*, 2024.
- **Hui Zhong**, Chenpei Huang, Xinyue Zhang, and Miao Pan. “Metaverse CAN: Embracing Continuous, Active, and Non-intrusive Biometric Authentication.” *IEEE Network*, 2023.
- Chenpei Huang, **Hui Zhong**, Jie Lian, Pavana Prakash, Dian Shi, Yuan Xu, and Miao Pan. “Eve Said Yes: AirBone Authentication for Head-Wearable Smart Voice Assistant.” *IEEE Transactions on Mobile Computing*, 2025.

Conference

- **Hui Zhong**, Xinyue Zhang, Hao Wang, Shucheng Yu, Yu Wang, and Miao Pan. “Enhanced Local Differential Privacy Protection against Crosstalk Attacks in Quantum Computing.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2025.

- **Hui Zhong**, Keyi Ju, Manojna Sistla, Xinyue Zhang, Xiaoqi Qin, Xin Fu, and Miao Pan. “Tuning Quantum Computing Privacy through Quantum Error Correction.” *IEEE Global Communications Conference (GLOBECOM)*, 2024.
- Chenpei Huang, **Hui Zhong**, Zhen Chen, Xiaoqi Qin, Lan Zhang, Xiaoyong Yuan, and Miao Pan. “EarID: In-Ear Wearable Device User Authentication Using Binary Acoustic Ear Canal Scanning Code.” *ACM Conference on Human Factors in Computing Systems (CHI) Late-Breaking Work*, 2025.
- Keyi Ju, **Hui Zhong**, Xinyue Zhang, Xiaoqi Qin, and Miao Pan. “Controllable Quantum Computing Privacy via Inherent Noises and Quantum Error Mitigation.” *IEEE International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, 2024.
- Yusheng Zhao, **Hui Zhong**, Xinyue Zhang, Yuqing Li, Chi Zhang, and Miao Pan. “Bridging Quantum Computing and Differential Privacy: Insights into Quantum Computing Privacy.” *IEEE International Conference on Quantum Computing and Engineering (QCE)*, 2024.
- Keyi Ju, Xiaoqi Qin, **Hui Zhong**, Xinyue Zhang, Miao Pan, and Baolong Liu. “Harnessing Inherent Noises for Privacy Preservation in Quantum Machine Learning.” *IEEE International Conference on Communications (ICC)*, 2024.
- Keyi Ju, Xiaoqi Qin, **Hui Zhong**, Xinyue Zhang, Miao Pan, and Baolong Liu. “Privacy Preserving Quantum Search Mechanism Using Grover’s Algorithm.” *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2024.
- Yuqing Li, Yusheng Zhao, Xinyue Zhang, **Hui Zhong**, Miao Pan, and Chi Zhang. “Differential Privacy Preserving Quantum Computing via Projection Operator Measurements.” *IEEE International Conference on Quantum Communications, Networking, and Computing (QCNC)*, 2024.
- Shushan Qiu, Yidan Shen, Yingpei Li, **Hui Zhong**, Zheng Wang, and Kaushik Rajasekhara. “Coordinated Optimal Energy Management Strategy for Grid-supporting Microgrid.” *IEEE Industrial Electronics Society (IECON)*, 2024.

OTHER PUBLICATIONS: UNDER REVIEW

Journal

- **Hui Zhong**, Keyi Ju, Manojna Sistla, Xinyue Zhang, Aohan Li, Xiaoqi Qin, Xin Fu, and Miao Pan. “Adjustable Quantum Computing Differential Privacy via Quantum Error Correction.” *Manuscript submitted, 2025*.
- **Hui Zhong**, Keyi Ju, Jiachen Shen, Xinyue Zhang, Xiaoqi Qin, Tomoaki Ohtsuki, Miao Pan, and Zhu Han. “Differential Privacy Preserving Quantum Distributed Computing.” *Manuscript submitted, 2025*.
- **Hui Zhong**, Jiachen Shen, Xinyue Zhang, Aohan Li, Beatriz Lorenzo, Miao Pan, and Zhu Han. “GARL: A Unified GNN–Attention and Reinforcement Learning Optimization Framework for Distributed Quantum Computing.” *Manuscript submitted, 2026*.

Conference

- **Hui Zhong**, Jiachen Shen, Xinyue Zhang, Aohan Li, Tomoaki Ohtsuki, Miao Pan, and Zhu Han. “Theoretical Modeling for Detecting Pulse Attacks in Quantum Circuits.” *Manuscript submitted, 2025*.
- **Hui Zhong**, Jiachen Shen, Lei Fan, Xinyue Zhang, Hao Wang, Miao Pan, and Zhu Han. “UNIQ: Communication-Efficient Distributed Quantum Computing via Unified Nonlinear Integer Programming.” *Manuscript submitted, 2025*.

TEACHING AND WORKING EXPERIENCE

Teaching assistant and Guest Lecturer

Jan. 2024 – Dec. 2024

Department of Electrical and Computer Engineering, University of Houston

Houston, TX, USA

- Courses: ECE 5397/6397 Introduction to Cybersecurity.
- Assisted in delivering lectures and discussions on core cybersecurity topics.
- Participated in the design and improvement of assignments and course materials.
- Answered student questions and helped deepen their understanding of cybersecurity fundamentals.

Software Development Intern

June. 2021 – Aug. 2021

Shanghai Zhongda Information Industry Co., LTD

Shanghai, China

- Assisted in software development by coding and debugging system functions.
- Collaborated with the team to design and implement customer-driven features.
- Improved system efficiency through optimized function execution.

Software Testing Intern

Jan. 2021 – Mar. 2021

Shanghai Shengben Intelligent Technology Co., LTD

Shanghai, China

- Executed functional and regression tests based on detailed test plans.
- Independently performed factory-level functionality verification for POS machines.
- Achieved 98% coverage and ensured high delivery quality.

PROFESSIONAL ACTIVITIES

Reviewer

- Journals: Future Generation Computer Systems, Journal of Air Transport Management, ACM Computing Surveys, IEEE Open Journal of Vehicular Technology, IEEE Journal on Selected Areas in Communications, ICT Express.
- Conferences: IEEE Global Communications Conference (GLOBECOM), IEEE Vehicular Technology Conference (VTC), International Conference on Artificial Intelligence, Information Processing and Cloud Computing (AIIPCC).

Memberships

- ACM Student Member since 2024

CONFERENCE PARTICIPATION

Oral Presentations

- Enhanced Local Differential Privacy Protection against Crosstalk Attacks in Quantum Computing, *IEEE International Conference on Quantum Computing and Engineering*, Albuquerque, United States, Sept. 2025.
- Bridging Quantum Computing and Differential Privacy: Insights into Quantum Computing Privacy, *IEEE International Conference on Quantum Computing and Engineering*, Quebec, Canada, Sept. 2024.
- Harnessing Inherent Noises for Privacy Preservation in Quantum Machine Learning, *IEEE International Conference on Communications*, Denver, United States, June. 2024.

Session Chair

- Toward Secure and Reliable Quantum Computing in the NISQ Era, *IEEE International Conference on Quantum Computing and Engineering*, Albuquerque, United States, Sept. 2025.

HONORS & AWARDS

- 2019 Mathematical Modeling Competition of Northeast China, Provincial First Prize
- 2020 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), Provincial First Prize
- 2020 10th MathorCup Mathematical Modeling Challenge, Undergraduate Division, Second Prize
- 2020 College Students' Innovation and Entrepreneurship Training Program, Provincial-level Project
- 2020–2021 Dalian Maritime University Outstanding Student Scholarship, Third Prize
- 2020–2021 Merit Student Award, Dalian Maritime University

REFERENCES

Contact 1:

Name: Miao Pan
Institution: University of Houston
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Phone Number: 713-743-0557
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Relationship: Ph.D. Advisor

Contact 3:

Name: Xinyue Zhang
Institution: Kennesaw State University
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Relationship: Research Collaborator

Contact 2:

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Institution: University of Houston
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Contact 4:

Name: Aohan Li
Institution: The University of Electro-Communications
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